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MIXING CONDITIONS, CENTRAL LIMIT THEOREMS, AND INVARIANCE PRINCIPLES: A SURVEY OF THE LITERATURE WITH SOME NEW RESULTS ON HETEROSCEDASTIC SEQUENCES

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□ *This article is a survey of the main results on the central limit theorem (CLT) and its invariance principle (IP) for mixing sequences that have been obtained in the probabilistic literature in the last fifty years or so with a view towards econometric applications. Each of these theorems specifies a set of moment, dependence, and heterogeneity conditions on the underlying sequence that ensures the validity of CLT and IP. Special emphasis is paid to the case in which the underlying sequence has just barely infinite variance, since this case is relevant to econometrics applications that involve high-frequency financial data. Moreover, two new results on IPs that apply to heteroscedastic sequences are obtained. The first IP applies to sequences whose variances evolve over time in a polynomial-like fashion, whereas the second IP concerns sequences that experience a single variance break at some point within the sample.*

Keywords Central limit theorem; Invariance Principle; Mixing; Trending variances; Variance break.

JEL Classification C22; C01.

1. INTRODUCTION

The advances in time series econometrics during the last 30 years or so have generated a considerable interest among econometricians on the various types of dependence in economic time series and the asymptotic theory under them. This interest stems from the desire to study the asymptotic properties of various estimators and test statistics that emerge in the context of linear and nonlinear models with non-independent innovations.

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The classic stochastic limit theory (CSLT) deals in large part with strong and weak limit theorems for partial sums of independent and identically distributed (i.i.d.) random variables. The strong limit theorems involve the concept of almost sure convergence and include results, such as the strong law of large numbers and the law of iterated logarithm. On the other hand, the weak limit theorems utilize the concept of convergence in distribution and include (mainly) results such as the central limit theorem (CLT). In this case, convergence in distribution is meant to imply convergence of probability measures defined on the Borel sets of the real line. Since the late 1940s, CSLT started to expand towards studying the convergence of a sequence of random functions constructed out of sums of i.i.d. random variables. The new results can be classified into two categories, which are usually referred to as strong and weak invariance principles (IPs). The weak IPs are theorems proving that the above-mentioned sequence of random functions converges in distribution to the standard Brownian motion (Donsker, 1951; Erdős and Kac, 1946). In this case, convergence in distribution refers to convergence of probability measures defined on the Borel sets of a function space. On the other hand, the strong IPs provide almost sure bounds on the order of magnitude of the error of approximation of the random functions by the standard Brownian motion (Strassen, 1964, 1965).

The modern stochastic limit theory (MSLT) has been a sustained effort towards extending the asymptotic results of CSLT for i.i.d. processes, to the case of non-i.i.d. ones. It might be said that the MSLT started in 1953 with the publication of the classic work *Stochastic Processes* by J. L. Doob, where the concept of martingale, introduced by Lévy (1935), was systematically investigated and turned into a powerful tool for probability theory. In this work, Doob obtained certain CLTs for martingales and Markov processes. At almost the same time, Marsaglia (1954) proved a CLT for m -dependent sequences of random variables, while a little later, Billingsley (1956) obtained the first weak IPs for Markov and m -dependent processes. At the same time, Prohorov (1956) obtained an IP for independent not necessarily identically distributed random variables. These generalizations of the classic results greatly increased the scope of applicability of CLTs and IPs and generated further interest in obtaining such results for more general modes of dependence.

Since then, the progress of MSLT has been rapid. This is mainly due to the development of various “operational” notions of dependence for stochastic processes. Rosenblatt (1956), Ibragimov (1959), and Kolmogorov and Rozanov (1960) introduced the concepts of strong mixing (α -mixing), uniform mixing (ϕ -mixing) and ρ -mixing, respectively, to characterize a stochastic process that is asymptotically independent. Some additional definitions of mixing have been proposed, such as ψ -mixing, or β -mixing (absolute regularity), all of which essentially require

the asymptotic decoupling of the “past” and the “future”. Bradley (2005) gives a comprehensive survey of these definitions and a brief discussion on the relationships among them.

CLTs and IPs for mixing sequences of random variables have extensively been employed in the recent econometric literature. For example, IPs constitute the starting point in the asymptotic analysis of estimators and test statistics applied to data generated by autoregressive processes with unit roots (see, for example, Phillips, 1987). In this article, we survey the main CLTs and IPs that have been proposed in the probabilistic literature since 1960, for three types of mixing sequences, namely, α -, ϕ -, and ρ -mixing.¹ Special emphasis is given to the results that do not assume finite second moments for the underlying sequence. These results are likely to be useful in the analysis of high-frequency financial data such as stock or exchange rate returns. Some of the results, presented in this article, may be found in other sources of the econometric literature, such as Davidson (1994) or White (2001). These studies, however, do not focus exclusively on mixing processes, but cover a broader spectrum of dependence structures, such as McLeish’s (1974; 1975; 1977) mixingale and the near epoch dependent (NED) ones (see also De Jong, 1997; Wooldridge and White, 1988). Since the literature on the general subject is vast, the aforementioned studies give only a brief account of the CLTs and IPs for mixing sequences that might be useful to econometrics. On the other hand, one can find excellent surveys of the subject in the probabilistic literature (see for example, the early surveys of Iosifescu, 1980 and Peligrad, 1986 or the more recent ones by Merlevede et al., 2006). However, these surveys have been written from a probability-theory perspective which may overemphasize aspects of the subject that are not of special interest to econometrics. For example, the techniques employed to solve the CLT problem for mixing sequences, such as Bernstein’s or Gordin’s methods of approximating mixing sequences by independent or martingale ones, respectively, usually attract more attention than the typical econometrician is willing to pay. As another example, the probabilistic reviews put special emphasis on the existence of necessary and sufficient conditions for CLTs and IPs, which are hard to verify or even interpret in practice. On the contrary, in econometrics, the interest typically centers on establishing sufficient conditions for such results, which are usually expressed as “regularity” conditions on the errors of an econometric model. Moreover, the technical issues for proving IPs, which are of fundamental importance in probability theory may not be of immediate and direct interest to econometrics. For example, the standard method for obtaining IPs is to first show that the finite-dimensional

¹The case of β -mixing (or absolute regularity) will also be mentioned, since it is often met in the econometric literature.

distributions of the suitably normalized partial sums, say $W_n(r)$, converge to the corresponding ones of Brownian motion and then to demonstrate that $W_n(r)$ is *tight* in the sense of Prohorov (1956). The conditions that establish the tightness property, which are of great interest to probabilists, do not offer much insight into the asymptotic problems typically faced by econometricians.

Based on the above mentioned considerations, this article aims to be of interest to econometricians by reviewing the literature on CLTs and IPs for mixing sequences alone. This is accomplished at the expense of omitting many useful limit theorems that employ types of dependence other than mixing. For example, apart from mixingales and NEDs, the concept of “associated” random variables, introduced by Esary et al. (1967), has led to certain IPs, in which the mixing conditions are replaced by a simple summability condition on the covariance (see, for example, Newman and Wright, 1981).

Special emphasis is given to the case of stochastic sequences $\{X_t, t \in \mathbb{N}\}$ with *just barely* infinite variances in the sense that $E|X_t|^\delta < \infty$, $0 \leq \delta < 2$. For this type of sequences, it has been proved that the CLT and IP hold (see Bradley, 1988; Peligrad, 1990; Shao, 1993). These results, which are not very well known to econometricians, are potentially useful to the case of modeling financial data. Indeed, this type of data is usually considered as realizations from an Integrated GARCH (IGARCH) process, which is a process with just barely infinite variance (see Kourougenis and Pittis, 2008). Moreover, apart from surveying the literature, this article offers some new results on IPs for heteroscedastic processes. In particular, we obtain IPs for two cases that are of special interest to econometrics: the first case assumes a process whose variance changes with time in a polynomial like fashion, whereas the second case deals with processes that experience a single variance shift at some point in the sample.

The article is organized as follows: Section 2 defines the asymptotic problems of interest and makes some preliminary remarks. Section 3 begins by presenting some results on necessary and sufficient conditions for the CLT for the case of mixing sequences with finite variances. It also contains references to various sources (including an extended version of this article) where the most important CLTs and IPs for the finite-variance case may be found. The largest part of Section 3 is devoted to surveying CLTs and IPs for the case in which the second moments of the underlying mixing sequence are “barely” infinite. Section 4 obtains modified IPs for the heteroscedastic processes mentioned above. Section 5 concludes the article.

2. DEFINITION OF THE PROBLEMS

Let $(\Omega, \mathcal{F}, P)$ be a probability space and $\{X_t, t \in \mathbb{N}\}$ be a sequence of random variables defined on this space. For the rest of the article, we

shall assume that $E(X_t) = 0$ for every t and use the symbols $N(0, 1)$ and W to denote the standard normal distribution and the standard Brownian motion on $[0, 1]$, respectively. Define the partial sum processes

$$S_n = \sum_{t=1}^n X_t.$$

The CLT deals with the following question: Under what conditions on the moment, dependence and heterogeneity properties of $\{X_t, t \in \mathbb{N}\}$, does there exist a sequence $\{b_n, n \geq 1\}$ of positive numbers with $\lim_{n \rightarrow \infty} b_n = \infty$ such that:

$$\frac{1}{b_n} S_n \xrightarrow{L} N(0, 1)? \quad (1)$$

If we can find a set of properties of $\{X_t, t \in \mathbb{N}\}$ for which the convergence result (1) is established, we say that $\{X_t, t \in \mathbb{N}\}$ obeys the CLT.

Next, for each $r \in [0, 1]$, set

$$S_n(r) = \sum_{t=1}^{[nr]} X_t,$$

where $[x]$ denotes the greatest integer not exceeding x . As before, assume that $\{b_n, n \geq 1\}$ is a (deterministic) sequence of positive real numbers such that $b_n \rightarrow \infty$ as $n \rightarrow \infty$, and define

$$W_n(r) = \frac{1}{b_n} S_n(r).$$

For each $n \geq 1$, consider the function W_n , defined on Ω , by $\omega \mapsto W_n(\cdot, \omega)$. Then W_n is a mapping from $(\Omega, \mathcal{F})$ to $(D([0, 1]), \mathcal{B}(D([0, 1])))$ where $D([0, 1])$ is the space of all functions defined on $[0, 1]$ which have left limits and are continuous from the right at every point (cadlag functions) and $\mathcal{B}(D([0, 1]))$ (or simply $\mathcal{B}$) is the Borel σ -algebra on $D([0, 1])$ induced by the Skorokhod topology. The mapping W_n is $\mathcal{F}/\mathcal{B}$ measurable in the sense that for each $n \geq 1$, and for every Borel set $B \in \mathcal{B}$, the pre-image of B under W_n is an event in $\mathcal{F}$. The IP deals with the following question: Under what conditions on the moment, dependence and heterogeneity properties of $\{X_t, t \in \mathbb{N}\}$, the sequence of random functions $\{W_n, n \geq 1\}$ converges in distribution to the standard Brownian motion, i.e.,

$$W_n \xrightarrow{L} W? \quad (2)$$

Remarks. (i) Both results (1) and (2) involve the same notion of convergence, which is that of “weak convergence of probability measures”

or “convergence in law.” In other words, both (1) and (2) state that a sequence of probability measures $\{P_n, n \geq 1\}$ converges weakly to a given probability measure, say P . However, (1) and (2) differ in one important respect: In the case of (1), P_n and P are defined on the Borel field $\mathcal{B}(\mathbb{R})$ of the set of real numbers. In particular, P_n are the probability measures induced by the random variables $\frac{1}{b_n}S_n$ on the measurable space $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ of the real line. This specific feature allows the probability measures (distributions) P_n and P to be replaced by the corresponding distribution functions, say F_n and F , respectively. In such a case, the concept of “weak convergence of probability measures” can be replaced by the more familiar concept of “pointwise convergence” of distribution functions. On the other hand, in the case of (2), P_n and P are defined on the Borel σ -algebra $\mathcal{B}(D([0, 1]))$. In particular, P_n are the probability distributions induced by the random functions W_n on the measurable space $(D([0, 1]), \mathcal{B}(D([0, 1])))$ (for an excellent discussion on these issues, see Billingsley, 1968).

(ii) As mentioned in the introduction, the standard method for obtaining IPs consists of two steps. In the first step we show that the finite-dimensional distributions of the W_n converge to those of W . This means that for all choices of $s_1, s_2, \dots, s_k$ and all $k \geq 1$, we must establish that

$$[W_n(s_1), W_n(s_2), \dots, W_n(s_k)] \xrightarrow{L} [W(s_1), W(s_2), \dots, W(s_k)]. \quad (3)$$

To establish (3), we may resort to classical methods such as those of characteristic functions. The second step is to prove that the sequence W_n is tight. By “tightness of W_n ,” we mean that the corresponding distributions, P_n , are tight. In general, a family of probability measures, Π on a metric space S is said to be tight if for every $\varepsilon > 0$ there exists a compact set K_ε such that $P(K_\varepsilon) > 1 - \varepsilon$ for every $P \in \Pi$. Intuitively, the tightness property prevents probability from “escaping to infinity” as we move through Π . For the case under study, sufficient conditions for the tightness of W_n can be found in Billingsley (1968, Theorems 15.2 and 15.3 p. 125).

(iii) The expressions (1) and (2) state the CLT and IP, respectively, in their full generality. Most of the existing literature, however, has focused attention on the case $b_n = \sigma_n$, where $\sigma_n^2 \equiv \text{Var}(S_n) = E(S_n^2)$, with $\lim_{n \rightarrow \infty} \sigma_n^2 = \infty$. A usual assumption of this literature is that the variances of the partial sums behave nicely. This assumption takes the form

$$\lim_{n \rightarrow \infty} E\left(\frac{S_n^2}{n}\right) = \sigma^2, \quad 0 < \sigma^2 < \infty \quad (4)$$

For example condition (4) is trivially satisfied for every uncorrelated stationary sequence. In such a case, the normalizing sequence, b_n , in (1)

and (2) is given by $\sigma\sqrt{n}$. This in turn implies that $\sqrt{n}$ is just the right rate at which we have to control S_n or $S_n(r)$ in order to obtain convergence to the normal distribution or the Brownian motion, respectively.

(iv) In the case that the stationary sequence $\{X_t, t \in \mathbb{N}\}$ approaches independence sufficiently fast, (a condition that usually takes the form of a restriction on the mixing rates) the parameter σ^2 takes the form,

$$\sigma^2 = E(X_1^2) + 2 \sum_{j=2}^{\infty} E(X_1 X_j). \quad (5)$$

This parameter, usually referred to as the “long-run variance” of $\{X_t, t \in \mathbb{N}\}$, summarizes the second-order properties of the sequence.

(v) In most of the cases that will be considered below, the normalizing sequence will be $\sigma\sqrt{n}$. However, there might be cases where (1) and (2) do not hold under this particular normalization, but they do hold under more general normalizing sequences.

3. CLTS AND IPS FOR MIXING SEQUENCES OF RANDOM VARIABLES

As mentioned above, a CLT or an IP involves a set of conditions on $\{X_t, t \in \mathbb{N}\}$ guaranteeing that (1) or (2), respectively holds. These conditions, however, although sufficient for CLT or IP, might not be necessary. The problem of finding necessary and sufficient conditions for the CLT in the case of α -mixing processes was initially dealt with by Ibragimov (1959, 1962). One of his necessary conditions refers to the behavior of the variance σ_n^2 of S_n .

Theorem 1 (A Necessary Condition for the CLT, Ibragimov, 1962). *Let $\{X_t, t \in \mathbb{N}\}$ be a strictly stationary, α -mixing sequence of random variables with $E(X_1) = 0$, $E(X_1^2) = \sigma_X^2 < \infty$. Then in order for $\frac{1}{\sigma_n} S_n \xrightarrow{L} N(0, 1)$ (and $\lim_{n \rightarrow \infty} \sigma_n^2 = \infty$), it is necessary that*

$$\sigma_n^2 = nh(n), \quad (6)$$

where $h(x)$ is a slowly varying function of the continuous variable $x > 0$.²

Remark. Condition (6) implies that σ_n^2 does not behave very differently from a linear function of n .

²A function L is slowly varying (at infinity) iff $\frac{L(tx)}{L(x)} \rightarrow 1$ as $x \rightarrow \infty$ for every $t > 0$.

The rest of Ibragimov's necessary conditions are rather complicated and refer to the rate at which the strong mixing coefficients approach zero, combined with a moment restriction. One of the simplest necessary and sufficient conditions for the CLT is the one obtained by Denker (1986).

Theorem 2 (A Necessary and Sufficient Condition for the CLT, Denker, 1986). *Let $\{X_t, t \in \mathbb{N}\}$ be a strictly stationary, α -mixing sequence of random variables with $E(X_1) = 0$, $E(X_1^2) = \sigma_X^2 < \infty$, and $\lim_{n \rightarrow \infty} \sigma_n^2 = \infty$. Then in order for $\frac{1}{\sigma_n} S_n \xrightarrow{L} N(0, 1)$, it is necessary and sufficient that the sequence $\{S_n^2/\sigma_n^2, n \geq 1\}$ is uniformly integrable.*

In other words, this theorem states that for strong mixing sequences, the CLT is equivalent to the uniform integrability of the sequence $\{S_n^2/\sigma_n^2, n \geq 1\}$. This condition is guaranteed if the sequence $\{E|S_n/\sigma_n|^{2+\delta}, n \geq 1\}$ for some $\delta > 0$, is bounded. Moreover, Peligrad (1985) showed that for ϕ -mixing sequences, with $\phi(1) < 1$, the uniform integrability condition is equivalent to the Lindeberg condition

$$\lim_{n \rightarrow \infty} \frac{n}{\sigma_n^2} EX_1^2 \mathbf{I}_{X_1^2 > \varepsilon \sigma_n^2} = 0, \quad \text{for all } \varepsilon > 0, \quad (7)$$

where $\mathbf{I}_A$ denotes the indicator of the set A , thus resulting in another necessary and sufficient condition for the CLT in the case of ϕ -mixing sequences.

Since the necessary and sufficient conditions, imposed on the moments of $|S_n|$ are difficult to verify, we turn to the problem of establishing reasonable sufficient conditions imposed on the moments of the original process $\{X_t, t \in \mathbb{N}\}$. In order for the theorems to be useful, the sufficient conditions must be “sharp,” in the sense that their violation allows the construction of counterexamples (see Merlevede and Peligrad, 2000). Since 1956, when Rosenblatt introduced the concept of α -mixing sequences, many authors have derived CLTs and IPs for mixing sequences of random variables. For example, Ibragimov (1962) proves that a strictly stationary α -mixing sequence with finite moments of order $2 + \delta$, $\delta > 0$ obeys a CLT, provided that the mixing coefficients, $\alpha(n)$, satisfy a power-type mixing rate, expressed as $\sum_{n=1}^{\infty} \alpha(n)^{\frac{\delta}{2+\delta}} < \infty$. Relaxing the stationarity assumption does not bring about any major complications (see Herrndorf, 1984). On the other hand, replacing the moment requirement $\delta > 0$ by $\delta \geq 0$ turned out to be a much more difficult problem (see Bradley, 1985; Gordin, 1969; Hall and Heyde, 1980; Herrndorf, 1983, 1985). A detailed description of the most important CLTs and IPs for α -, ϕ -, and ρ -mixing sequences with finite second moments that have been obtained since 1956 can be found in an extended version of this article (henceforth KP2) available at the Web

site http://papers.ssrn.com/sol3/papers.cfm?abstract_id=1331319 (see also Hall and Heyde, 1980; Ibragimov and Linnik, 1971; Peligrad, 1986). There, the problem of finding a set of conditions that are “minimally” (or “barely”) sufficient for the CLT is also analyzed in some detail (see, for example, Doukhan et al., 1994). In the current version of the article, we focus mainly on certain more recent results for the case in which the mixing sequences under study have infinite second moments.

3.1. CLTs and IPs for Mixing Sequences with Infinite Second Moments

The question that we address in this section is the following: If the variance of (the identically distributed) X_t is assumed to be infinite, then what can be said about the weak convergence of the properly normalized S_n (and that of $W_n(r)$)? The answer to this question must take into account the following two issues. First, in the absence of finite variances of X_t , the standardizing sequence b_n cannot be the sequence of standard deviations σ_n . Second, the limiting distribution need not be the standard normal.

Historically, the problem described above was first dealt with by Lévy (1935) in the context of independent and identically distributed (*iid*) random variables. Specifically, Lévy asked (and answered) the following two questions: (i) What is the family, $\mathcal{L}$, of *all possible limit laws* of normalized sums of *iid* random variables? (ii) For a specific limit law of this family, can we find the corresponding conditions that ensure convergence to this law? Lévy showed that $\mathcal{L}$ is the family of the so-called “stable laws,” whose members are characterized by a parameter, a , $0 < a \leq 2$. This parameter is called the *index* or *characteristic exponent* and for the case of the normal distribution, $a = 2$. Ibragimov and Linnik (1971) showed that the family of all possible limit laws is still $\mathcal{L}$ when the assumption of independence is replaced by that of mixing.

Before we proceed, let us introduce some terminology and preliminary results which were initially developed in the context of independence, but apply equally well to the case of mixing. Assume that $\{X_t, t \in \mathbb{N}\}$ is an *iid* sequence of random variables with marginal distribution function, $F(x)$. Moreover, suppose that there exists two sequences $\{b_n, n \geq 1\}$ and $\{\gamma_n, n \geq 1\}$ of positive and real numbers respectively such that

$$\frac{1}{b_n} S_n - n\gamma_n$$

converges weakly to a random variable Y with distribution function $G_a(y) \in \mathcal{L}$ with exponent a . Then F (or the sequence $\{X_t, t \in \mathbb{N}\}$) is said to *belong to the domain of attraction (DA) of the stable distribution function G_a with*

exponent a . Feller (1971) derives necessary and sufficient conditions for F to belong to the DA of G_a . First, define the truncated moment function

$$H(x) = EX_1^2 \mathbf{I}_{|X_1| \leq x}.$$

For the case that $a < 2$, F belongs to the DA of G_a if and only if

$$H(x) \sim x^{2-a} L(x), \quad (8)$$

where $L(x)$ is a slowly varying function at infinity, and

$$\lim_{x \rightarrow \infty} \frac{1 - F(x)}{1 - F(x) + F(-x)} = p, \quad (9)$$

$$\lim_{x \rightarrow \infty} \frac{F(-x)}{1 - F(x) + F(-x)} = q, \quad (10)$$

where $0 \leq p \leq 1$ and $q = 1 - p$. Condition (8) states that the truncated moment function varies regularly with an exponent $2 - a$. On the other hand, condition (9) implies that the two tails of F are similarly balanced.

Assume that the distribution function $F(x)$ belongs to the DA of a stable law with exponent a . From this fact, can we infer anything about the existence of moments of the summands X_i ? Khintchine (1935), Cramér (1937), and Gnedenko (1949) provided an affirmative answer to this question. Specifically they showed that for every δ such that $0 \leq \delta < a$, the moment $E|X|^\delta$ exists. Moreover, if $a < 2$, no moments of order $\delta > a$ exist. This in turn implies that in order for a normalized sum of *i.i.d.* random variables to converge to the normal distribution ($a = 2$), the assumption of finite variance is not necessary. What is required, instead, is the truncated moment function, $H(x)$ to be slowly varying as $x \rightarrow \infty$, that is

$$H(x) \text{ is slowly varying as } x \rightarrow \infty \quad (11)$$

(see (8) for $a = 2$). In fact, the condition of slow variation of $H(x)$ is both necessary and sufficient for F to lie in the DA of the normal distribution.

Let us now turn our attention to the determination of the normalizing sequence $\{b_n, n \geq 1\}$. If F belongs to the DA of G_a , then it is certain that the normalizing sequences $\{b_n, n \geq 1\}$ and $\{\gamma_n, n \geq 1\}$ such that $\frac{1}{b_n} S_n - n\gamma_n \xrightarrow{L} G_a$ do exist. The next obvious question is how these sequences are determined. In proving the equivalence of (8) and (9) to $F \in DA(G_a)$, Feller (1971) obtains the following condition that $\{b_n, n \geq 1\}$ is required to satisfy

$$\lim_{n \rightarrow \infty} \frac{nH(b_n)}{b_n^2} = C. \quad (12)$$

This condition may be used to determine $\{b_n, n \geq 1\}$. In the case that the latter is of the form $b_n = n^{\frac{1}{a}}$, then the attraction is called “normal” and F is said to belong to the *domain of the normal attraction (DNA)* of G_a . This terminology was introduced by Gnedenko (1949) and must not be confused with the concept of the “domain of attraction to the normal distribution.”

So far, the assumption of independence of $\{X_t, t \in \mathbb{N}\}$ was implicit in the discussion. We shall now address the question of whether a mixing sequence of random variables can belong to the *DA* of the normal distribution, even if the random variables do not possess second-moments. In other words, can we obtain CLTs and IPs if we relax the assumption of “finite variance?” The answer is in the affirmative. The key assumption for obtaining such results is again (11). To this end, Bradley (1988) obtained the following theorem.

Theorem 3 (Bradley, 1988). *Let $\{X_t, t \in \mathbb{N}\}$ be a strictly stationary, ρ -mixing sequence of random variables with $E(X_1) = 0$, $\rho(1) < 1$, satisfying the conditions (11) and*

$$\sum_{i=1}^{\infty} \rho(2^i) < \infty. \quad (13)$$

Then there exists a sequence $\{b_n, n \geq 1\}$ of positive numbers with $\lim_{n \rightarrow \infty} b_n = \infty$ such that

$$\frac{1}{b_n} S_n \xrightarrow{L} N(0, 1). \quad (14)$$

Remark. Condition (11) replaces the usual but stronger condition $E(X_1^2) < \infty$. This assumption implies that $E|X_1|^\delta < \infty$ for every δ , $0 \leq \delta < 2$ which means that $\{X_t, t \in \mathbb{N}\}$ has *just barely* infinite second moments. Without assumption (11), the weak convergence of $\frac{1}{b_n} S_n$ to $N(0, 1)$ cannot be obtained.

Shao (1993) extended the above CLT to an Invariance Principle.

Theorem 4 (Shao, 1993). *Under the conditions of Theorem 3, we have that*

$$W_n(r) = \frac{1}{b_n} S_n(r) \xrightarrow{L} W.$$

Condition (11) is obviously satisfied if $E(X_1^2) < \infty$. Furthermore, Ibragimov and Linnik (1971) proved that (11) is satisfied if $P(|X_1| > x)$ is

regularly varying with the exponent -2 , that is

$$P(|X_1| > x) = x^{-2}L(x), \quad (15)$$

where $L(x)$ is, as usual, a slowly varying function at infinity.

Peligrad (1990) investigated the problem of obtaining invariance principles for ϕ -mixing sequences with infinite variance. In particular, she asked the question of whether the so-called “Iosifescu conjecture” can be extended to the case of a ϕ -mixing sequence with infinite variance that satisfies condition (11).³ Peligrad failed to validate this conjecture in its full generality. However, she succeeded in proving that this conjecture is true under the stronger condition (15). This result is stated in the form of the following theorem.

Theorem 5 (Peligrad, 1990). *Let $\{X_t, t \in \mathbb{N}\}$ be a strictly stationary, ϕ -mixing sequence of random variables with $E(X_1) = 0$, $\phi(1) < 1$, satisfying condition (15). Then there exists a sequence $\{b_n, n \geq 1\}$ of positive numbers with $\lim_{n \rightarrow \infty} b_n = \infty$ such that*

$$W_n(r) = \frac{1}{b_n} S_n(r) \xrightarrow{L} W.$$

Remark. Peligrad’s result, as opposed to that of Shao (1993), does not require the mixing rates to decay at a specific rate. However, her result is more restrictive than Shao’s in the sense that it is proved under condition (15) which is stronger than (11).

An interesting econometric application of the CLTs and IPs for sequences with just barely infinite variance is the case of Integrated GARCH(1, 1) (IGARCH) sequences. These processes are commonly used to describe financial time series that exhibit a high degree of volatility persistence as well as excess kurtosis. The preceding analysis implies that an IGARCH(1, 1) process may obey a CLT, or even an IP, despite the fact that the variance of this process is infinite. This is due to the fact that an IGARCH(1, 1) has just barely infinite variance. Indeed, based on Nelson’s (1990) results, it is easy to show that the absolute moments of order $\delta < 2$ for an IGARCH(1, 1) process exist. Moreover, such a process is strictly stationary and β -mixing with exponentially decaying mixing rates (see Francq and Zakoian, 2006). However, we cannot claim that a strictly stationary IGARCH(1, 1) process necessarily satisfies the conditions of

³Iosifescu (1977) conjectured that we may obtain an IP for strictly stationary, ϕ -mixing sequences, assuming only $E(X_1) = 0$, $E(X_1^2) < \infty$ and $\lim_{n \rightarrow \infty} \sigma_n^2 = \infty$ (see Section 3 in KP2 for a description of the Iosifescu conjecture).

Theorem 4 or those of Theorem 5, since there is no proof in the literature that an IGARCH process is either ϕ -mixing or ρ -mixing. Nevertheless, the above mentioned results allow us to conjecture with a reasonable degree of confidence that a martingale difference IGARCH(1, 1) sequence obeys the invariance principle.

The fact that there exist mixing sequences with barely infinite variance which obey the standard IP (as stated by Theorems 4 and 5) was utilized by Kourrogenis and Pittis (2008) in the context of unit root testing. Specifically, this study assumes that the variance of the innovations (driving the unit root process) is barely infinite and derives a necessary and sufficient condition under which the standard Z-statistics, proposed by Phillips (1987), converge to the corresponding Dickey–Fuller distributions. This in turn implies that, under the derived condition, the unit root hypothesis can be tested within an infinite-variance framework without any modifications to the existing testing strategies. Indeed, the simulation results reported in the aforementioned study suggest that the standard unit root tests exhibit very good properties in the context of a unit-root process driven by IGARCH(1, 1) innovations.

4. MODIFIED INVARIANCE PRINCIPLES FOR HETEROSCEDASTIC PROCESSES

The theory presented in the previous sections relied on certain stationarity assumptions on the stochastic sequence $\{X_t, t \in \mathbb{N}\}$. In the econometrics literature, we often deal with stochastic sequences for which the stationarity assumption is clearly inappropriate. In this section, we study three types of heterogeneity for the sequence $\{X_t, t \in \mathbb{N}\}$, all of them assume nonconstant variances. The first one assumes that the variances of the X_t 's evolve over time in a polynomial-like fashion. The second type of heteroscedasticity considers a sudden change (break) in the variance of the sequence from one level to another. In both cases the main assumption is that a generic stochastic sequence satisfies the IP. The third type restricts significantly the generic stochastic sequence, allowing on the other hand for a general type of heterogeneity. The modified IPs, obtained in the following sections may prove to be useful in many cases of interest to econometrics such as testing for unit roots or cointegration in models with unconditionally heteroscedastic errors.

4.1. The Case of Trending Variances

Let us consider the following set of assumptions for the sequence $\{X_t, t \in \mathbb{N}\}$:

Assumption U. Let $\{X_t, t \in \mathbb{N}\}$ be a stochastic sequence such that $X_t = \sqrt{f(t)}v_t$, where $f(t) = \beta_k t^k + g(t) > 0$ with $\beta_k > 0$, $g(t) = O(t^{k-1})$ if $k \geq 1$, and $g(t) = 0$ if $0 \leq k < 1$, where $\{v_t\}_{t \geq 1}$ is a stochastic process, for which the IP holds with normalizing sequence $\sigma_{v,n}$, $\sigma_{v,n} \rightarrow \infty$ as $n \rightarrow \infty$, i.e.,

$$W_{v,n}(r) = \frac{1}{\sigma_{v,n}} S_{v,n}(r) \xrightarrow{L} W(r),$$

where $S_{v,n}(r)$ is the partial sums process defined by the v_t 's and $W(\cdot)$ is a standard Brownian Motion.

Then the following theorem holds.

Theorem M₁. Let $\{X_t, t \in \mathbb{N}\}$ be a stochastic sequence, for which Assumption U holds, then

$$\begin{aligned} Q_{k,n}(r) &= \frac{1}{n^{k/2} \sigma_{v,n}} S_n(r) \xrightarrow{L} Q = r^{\frac{k}{2}} W(r) - \frac{k}{2} \int_0^r s^{\frac{k}{2}-1} W(s) ds \\ &= \int_0^r s^{\frac{k}{2}} dW(s). \end{aligned}$$

Proof. First note that without any loss of generality we can assume that $\beta_k = 1$, since we can set $v_t^* = \beta_k v_t$, $g^*(t) = \frac{g(t)}{\beta_k}$ and $\sigma_{v,n}^* = \beta_k \sigma_{v,n}$. Then

$$\frac{1}{\sigma_{v,n}^*} S_{v,n}^*(r) \xrightarrow{L} W(r),$$

where $S_{v,n}^*(r)$ is the partial sums process defined by the v_t^* 's.

We observe that

$$\begin{aligned} \sigma_{v,n}^{-1} S_n(r) &= \sigma_{v,n}^{-1} \sum_{i=1}^{[nr]} X_i = \sigma_{v,n}^{-1} \sum_{i=1}^{[nr]} \sqrt{f(i)} v_i \\ &= \sigma_{v,n}^{-1} \sum_{i=1}^{[nr]} i^{\frac{k}{2}} \int_{\frac{i-1}{n}}^{\frac{i}{n}} dS_{v,n}(s) + \sigma_{v,n}^{-1} \sum_{i=1}^{[nr]} O(i^{\frac{k}{2}-1}) v_i \\ &= \sigma_{v,n}^{-1} n^{\frac{k}{2}} \sum_{i=1}^{[nr]} \int_{\frac{i-1}{n}}^{\frac{i}{n}} s^{\frac{k}{2}} dS_{v,n}(s) + \sigma_{v,n}^{-1} \sum_{i=1}^{[nr]} O(i^{\frac{k}{2}-1}) v_i \\ &= \sigma_{v,n}^{-1} n^{\frac{k}{2}} \int_0^{\frac{[nr]}{n}} s^{\frac{k}{2}} dS_{v,n}(s) + \sigma_{v,n}^{-1} \sum_{i=1}^{[nr]} O(i^{\frac{k}{2}-1}) v_i \end{aligned}$$

$$\begin{aligned}
\Rightarrow n^{-\frac{k}{2}} \sigma_{v,n}^{-1} S_n(r) &= \int_0^r s^{\frac{k}{2}} dW_{v,n}(s) - \int_{\lfloor \frac{nr}{n} \rfloor}^r s^{\frac{k}{2}} dW_{v,n}(s) \\
&\quad + n^{-\frac{k}{2}} \sigma_{v,n}^{-1} \sum_{i=1}^{\lfloor nr \rfloor} O(i^{\frac{k}{2}-1}) v_i.
\end{aligned} \tag{16}$$

Since every realization of $W_{v,n}$ is a simple function, we have that

$$\int_{\lfloor \frac{nr}{n} \rfloor}^r s^{\frac{k}{2}} dW_{v,n}(s) = 0.$$

Moreover, since $s^{\frac{k}{2}}$ is continuous, we can integrate by parts the integral of the first term of the right-hand side of (16) to obtain

$$\int_0^r s^{\frac{k}{2}} dW_{v,n}(s) = r^{\frac{k}{2}} W_{v,n}(r) - \frac{k}{2} \int_0^r s^{\frac{k}{2}-1} W_{v,n}(s) ds.$$

Applying the continuous mapping theorem and the fact that $W_{v,n}(r) \xrightarrow{L} W(r)$, we finally obtain

$$\int_0^r s^{\frac{k}{2}} dW_{v,n}(s) \xrightarrow{L} r^{\frac{k}{2}} W(r) - \frac{k}{2} \int_0^r s^{\frac{k}{2}-1} W(s) ds. \tag{17}$$

Using (17) along with the fact that $\sigma_{v,n} \rightarrow \infty$ as $n \rightarrow \infty$ in (16), we conclude that

$$Q_{n,k}(r) = n^{-\frac{k}{2}} \sigma_{v,n}^{-1} S_n(r) \xrightarrow{L} r^{\frac{k}{2}} W(r) - \frac{k}{2} \int_0^r s^{\frac{k}{2}-1} W(s) ds = Q(r). \tag{18}$$

$Q(\cdot)$ involves a Brownian motion and a Riemann integral. By applying Itô's formula, $Q(\cdot)$ may be expressed in terms of a stochastic integral of a power function over a Brownian motion. Specifically,

$$\begin{aligned}
dQ(r) &= d(r^{k/2} W(r)) - \frac{k}{2} r^{k/2-1} W(r) dr \\
&= r^{k/2} dW(r) + \frac{k}{2} r^{k/2-1} W(r) dr - \frac{k}{2} r^{k/2-1} W(r) dr \\
&= r^{k/2} dW(r).
\end{aligned}$$

Therefore,

$$Q(r) = \int_0^r s^{\frac{k}{2}} dW(s).$$

Remark. We can easily see that when $k = 0$, $Q(\cdot)$ coincides with $W(\cdot)$, which in turn implies that Assumption U renders the existing results as special cases. When $k > 0$ the structure of $Q(\cdot)$ is more complex than that for the trivial case $k = 0$. In fact, the last expression of $Q(\cdot)$ and the Itô isometry imply that for every $k \geq 0$, $Q(\cdot)$ is a martingale with

$$Q(r) \sim N\left(0, \frac{r^{k+1}}{k+1}\right) \quad \text{for every } r \geq 0.$$

The type of unconditional heteroscedasticity defined by Assumption U may describe several financial time series. For example, Campbell et al. (2001) suggest that the idiosyncratic component of the unconditional variance of the returns of individual stocks exhibits a large positive linear trend over a 35-year period. The presence of such a trend dominates the behavior of the total firm volatility, thus producing a returns process with trending variance.

4.2. The Case of a Variance Break

Let us investigate now the case where the heterogeneity of the X_t 's stems from a variance break. Concerning the sequence $\{X_t, t \in \mathbb{N}\}$, we make the following assumption.

Assumption A. $\{X_t, t \in \mathbb{N}\}$, $\{v_t, t \in \mathbb{N}\}$ are two stochastic sequences such that

$$X_t = v_t, \quad 1 \leq t < [sT]$$

and

$$X_t = \sqrt{a}v_t, \quad [sT] \leq t \leq T,$$

where $s \in (0, 1]$, $a > 0$, and

$$\frac{1}{b_n} S_{v,n}(r) = \frac{1}{b_n} \sum_{t=1}^{[rn]} v_t \xrightarrow{L} W_1(r), \quad 0 \leq r \leq 1,$$

where $W_1(\cdot)$ is a standard Brownian Motion and

$$\frac{b_{[rn]}}{b_n} \rightarrow \sqrt{r}.$$

In order to describe the properties of the limiting distribution in the next theorem, we define the following function.

Definition. $D(\cdot)$ is a function defined on the interval $[0, 1]$, such that

$$\begin{aligned} D(r) &= 1 \quad \text{for } r < s \quad \text{and} \\ D(r) &= \sqrt{a + \frac{s(1-a)}{r}} \quad \text{for } r \geq s. \end{aligned}$$

Let us set $S_n(r) = \sum_{t=1}^{[rn]} X_t$ for $\frac{1}{n} \leq r \leq 1$ and $S_n(r) = 0$ for $0 \leq r \leq \frac{1}{n}$. Then, the following theorem holds.

Theorem M₂. *If Assumption A holds, then*

$$W_n(r) = \frac{1}{b_n} S_n(r) \xrightarrow{L} V_{a,s}(r) \equiv W_1(r \wedge s) + \sqrt{a} W_2((r - s) \vee 0), \quad (19)$$

where $W_1(\cdot)$ and $W_2(\cdot)$ are two independent standard Brownian Motions.

Proof. For $0 < r < s$, we have

$$\begin{aligned} \frac{1}{b_n} S_n(r) &= \frac{1}{b_n} \sum_{t=1}^{[rn]} X_t = \frac{1}{b_n} \sum_{t=1}^{[rn]} v_t, \\ \frac{b_{[rn]}}{b_n} \frac{1}{b_{[rn]}} \sum_{t=1}^{[rn]} v_t &\xrightarrow{L} \sqrt{r} N(0, 1) \equiv N(0, r). \end{aligned} \quad (20)$$

For $s \leq r \leq 1$, we also have

$$\begin{aligned} \frac{1}{b_n} S_n(r) &= \frac{1}{b_n} \sum_{t=1}^{[sn]-1} v_t + \frac{\sqrt{a}}{b_n} \sum_{t=[sn]}^{[rn]} v_t \\ &= \frac{b_{[sn]-1}}{b_n} \frac{1}{b_{[rn]-1}} \sum_{t=1}^{[sn]-1} v_t + \frac{b_{[rn]-[sn]+1}}{b_n} \frac{\sqrt{a}}{b_{[rn]-[sn]+1}} \sum_{t=[sn]}^{[rn]} v_t \\ &\xrightarrow{L} \sqrt{s} N_1(0, 1) + \sqrt{a} \sqrt{r-s} N_2(0, 1) \\ &\equiv N_1(0, s) + \sqrt{a} N_2(0, r-s), \end{aligned} \quad (21)$$

where N_1 and N_2 are normal independent. By virtue of Assumption A and equations (20) and (21), the proof is complete.

Remarks. (i) We observe that the process $S_n(\cdot)$ divided only by the sequence b_n does not converge to a standard Brownian Motion. Nevertheless, $V_{a,s}(\cdot)$ is zero mean Gaussian with almost surely continuous sample paths, independent increments and for every $r \in [0, 1]$,

$$E[V_{a,s}^2(r)] = rD^2(r).$$

Therefore,

$$V_{a,s}(r) \sim N(0, rD^2(r)) \quad \text{for every } r \geq 0.$$

(ii) The previous remark implies that the process $V_{a,s}(\cdot)$ cannot be transformed to a standard Brownian Motion. To see this, observe that if, for every $r \in [0, 1]$, we divide $S_n(r)$ by $b_n D(r)$, then the limiting distribution is $\frac{1}{D(r)} V_{a,s}(r)$, which is Gaussian with variance equal to r , but the process $\frac{1}{D(\cdot)} V_{a,s}(\cdot)$ has not independent increments.

4.3. A General Case of Time Heterogeneity

In the previous cases, time heterogeneity was introduced by multiplying a generic stochastic sequence $\{v_t\}_{t \geq 1}$ by a deterministic function, say σ_t . The only assumption on v_t s was that they satisfy the I. P. Cavaliere (2004) proved that if we restrict ourselves to the case of a strictly stationary sequence $\{v_t\}_{t \geq 1}$, then under additional assumptions we can obtain a modified IP for a general type of σ_t . The result is stated in the next theorem.

Theorem 6 (Cavaliere, 2004). *Let $X_t = \sigma_t v_t$ where $\{v_t\}$ is a zero mean, strictly stationary α -mixing process with $Ev_t^2 = 1$, $E|v_t|^p < \infty$ for some $p > 2$ and with mixing coefficients $\{a_m\}$ satisfying $\sum_{m=0}^{\infty} a_m^{2(1/r-1/p)} < \infty$ for some $r \in (2, 4]$, $r \leq p$. Furthermore, the long run variance $\lambda_\varepsilon^2 = \sum_{k=-\infty}^{\infty} E(v_t v_{t+k})$ is strictly positive and finite and $\sigma_{[sT]} = \omega(s)$, where $\omega(\cdot)$ is strictly positive, non-stochastic with finite number of points of discontinuity satisfying a first-order Lipschitz condition except at points of discontinuity. Then*

$$\frac{1}{\sqrt{n\bar{\sigma}\lambda_\varepsilon}} S_n(r) \xrightarrow{L} B_\omega(r), \quad \text{as } t \rightarrow \infty,$$

where $\bar{\sigma}^2 = \int_0^1 \omega(r)^2 dr$ and $B_\omega(s) = \bar{\sigma}^{-1} \int_0^s \omega(r) dW(r)$.

Remarks. (i) The above theorem assumes a type of unconditional heteroscedasticity which is more general than those covered by Assumptions U and A . This, however, comes at a price in the form of stricter assumptions on the generic process $\{v_t\}$. More specifically, this theorem assumes that $\{v_t\}$ is a strictly stationary process with finite moments of order $p > 2$. On the other hand, the only property that is required from $\{v_t\}$, in the context of assumptions U and A , is to satisfy the Invariance Principle. This in turn implies that assumptions U and A allow $\{v_t\}$ to be (mildly) heteroscedastic, or to have barely infinite second moments.

(ii) When the process exhibits heteroscedasticity of a general form, as the one described in Theorem 6, we can divide the partial sums process by the nonconstant function $\lambda_\varepsilon \sqrt{\int_0^r \omega(s)^2 ds}$, to obtain

$$\frac{1}{\sqrt{n} \lambda_\varepsilon \sqrt{\int_0^r \omega(s)^2 ds}} S_n(r) \xrightarrow{L} V(r), \quad \text{as } t \rightarrow \infty,$$

where $V(r)$ has all the properties of the standard Brownian motion except from the fact that its increments are not independent. This in turn implies that in contrast with the homoscedastic cases, the standardizing weights used in this modified IP are different from those used in the corresponding CLT.

5. CONCLUSIONS

This article has reviewed the central limit theorems and invariance principles for mixing sequences of random variables that have been obtained in the probabilistic literature since 1960. All the theorems surveyed in this article consist of sufficient conditions imposed on the moment, dependence, and heterogeneity properties of the underlying sequence. Special emphasis was given to theorems that ensure the validity of CLTs and IPs for mixing sequences with just barely infinite variances. These results are likely to be useful in many situations of practical interest to econometrics, especially those dealing with high-frequency financial data. These results, collected and discussed in one place, may enhance our understanding of the modern stochastic limit theory that is needed for the derivation of asymptotic results in the current econometrics literature. The survey has focused exclusively on mixing sequences of random variables without entering the vast domain of alternative dependence structures that have been recently suggested in the literature, such as those that employ innovative martingale-like conditions. To this end, a recent article by Merlevede et al. (2006) which surveys some recent results on CLTs and IPs under more general structures of dependence is likely to be of some interest. Finally, the article has offered two new results on IPs that apply to heteroscedastic sequences that exhibit either polynomial-like trending variances or variance breaks.

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